verdant.

Watering you can see and control.

Arduino automation with a dashboard for your phone.

A care routine leaves unanswered questions.

“Does it need water?”

Soil condition and reservoir supply need separate checks.

“What needs attention?”

Separate checks make it harder to see the growing conditions together.

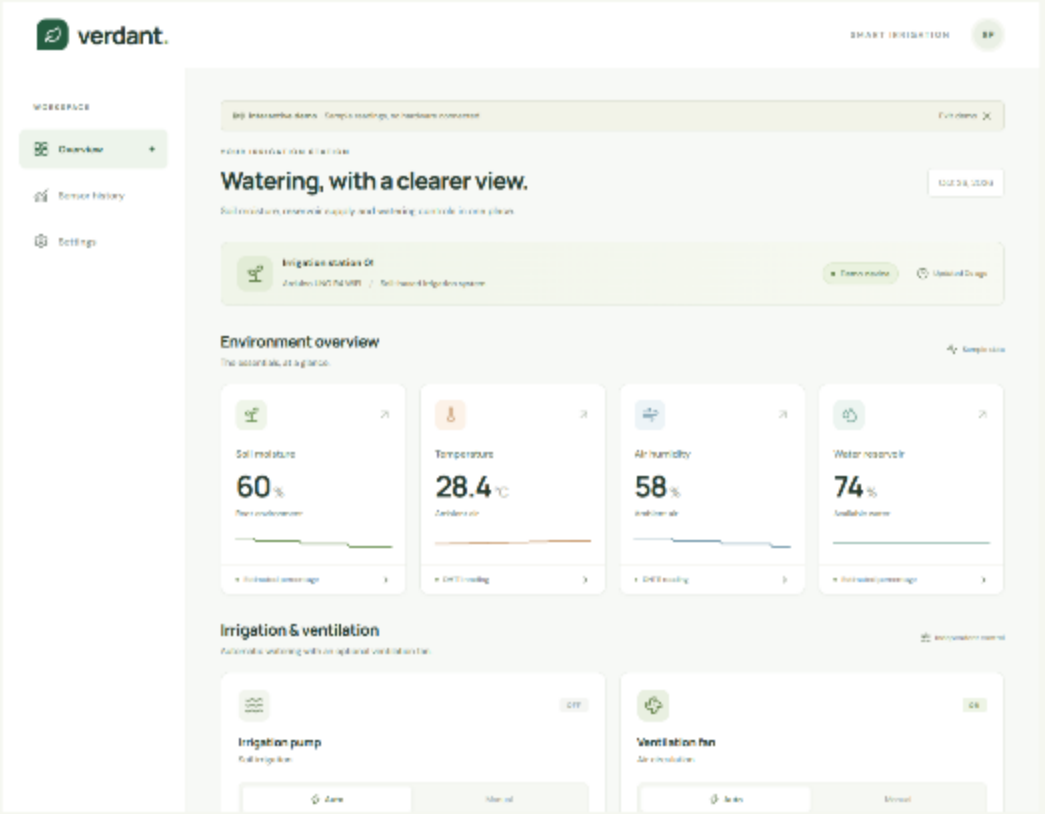
The opportunity: put the readings and the response in one place.

Soil moisture and water supply guide irrigation

Soil moisture
Reservoir level

Air conditions for context

Automatic or timed watering.
Optional ventilation control.



Actual dashboard with simulator data

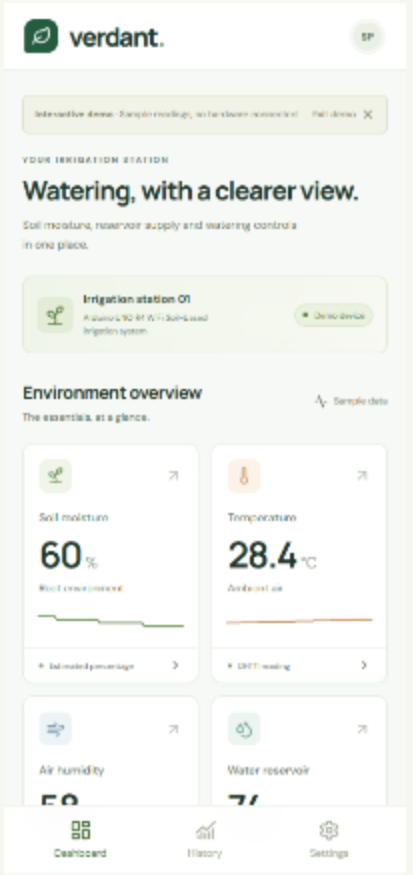
A phone becomes the control panel.

Check conditions

Read current values and reported output states.

Choose the response

Keep soil-based Auto, or run the pump for a limited time.



Simulator screen

One Arduino connects sensing and action.

Component	Role	Connection
Soil + reservoir sensors	Measure moisture and available water	A0 + A1
DHT11	Measure temperature and humidity	A2
Ventilation fan	Optional air circulation	D5 · MOSFET
Irrigation pump	Water the soil	D8 · relay
OLED display	Show readings locally	I²C · 0x3C

Local Wi-Fi keeps the demo simple.

01 Arduino

Reads sensors, runs Auto rules, applies outputs and timed commands.

02 Laptop

Serves the React dashboard and forwards local HTTP requests.

03 Phone

Shows readings and sends control requests through the browser.

Same local Wi-Fi. No internet or database needed during the demo.

The laptop stays on. Automatic control runs on the Arduino.

Soil-based watering with manual override

Auto

Pump ON: soil ≤30% and water ≥25%.
Pump OFF: soil ≥45% or water ≤15%.

Manual

30 seconds, 1 minute or 5 minutes. The board ends the timer and returns to Auto.

Built-in limits

Existing protective stops remain active. Stale readings disable new controls.
The board reports output state without separate physical actuator feedback.

Watering visibility for small growing spaces

For a plant owner

One view replaces separate checks. A timed action avoids leaving a manual activation running indefinitely.

For a classroom

Sensor readings connect directly to observable control decisions in a mobile interface.

Initial audiences to test, not confirmed buyers.

An irrigation kit with guided setup

The proposed offer

A proposed irrigation kit with a pump, soil and water sensors, and dashboard setup.

The first sales test

Show the prototype to indoor plant owners and teaching labs. Test usefulness and willingness to pay.

Price follows a bill of materials and pilot feedback.

The software works in simulation. Hardware checks come next.

Completed software checks

19 automated tests passed.
Frontend built. Companion sketch compiled.
Dashboard and simulator exchanged data.

Before the physical demonstration

Confirm real sensor readings and outputs.
Check protective stops and disconnects.
Verify phone access over the hotspot.

Calibration remains untested. Soil and reservoir percentages are estimates.

The demonstration makes the product tangible.

01	**Connect**	Join the demo Wi-Fi and open the laptop dashboard URL.
02	**Observe**	Show the four sensor readings and actuator states.
03	**Activate**	Select the pump, then run a 30-second watering action.
04	**Return**	Watch the board timer finish and Auto resume.

Verdant makes watering visible

See the soil.
Control the watering.

Try the prototype. Help shape the first irrigation kit.

Next: physical integration checks, then a small user pilot.